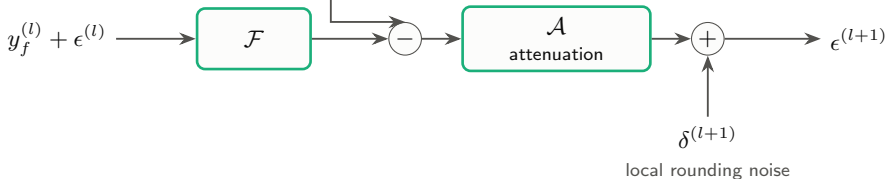


FP32 reference



quantized engine



Measured per node n :
$$a_n = \frac{r_{\text{out}}}{\max r_{\text{in}}}, \quad r = \|q - f\|_2 / \|f\|_2$$

$a_n < 1$: the operator attenuates propagated deviation (e.g. max-pooling of non-maximal noise);

$a_n > 1$: amplification, or fresh rounding noise δ injected by a quantized operator.